

ORIGINAL ARTICLE

Quantitative Assessment of *Ailia coila* (Hamilton, 1822) Fish Population in Kaptai Lake: A Length-Based Approach

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ABSTRACT

The *Ailia coila*, a small indigenous freshwater fish species in Kaptai Lake, plays a crucial role in providing employment, food, and financial security for the local people. With a high market demand, the average production of this species over the last 5 years reached about 112.27 metric tons. However, there has been a notable decline in *Ailia coila* fish production in Kaptai Lake in recent years. Therefore, to understand the current stock status of this species, this study employed a length-based stock assessment method the “length-based spawning potential ratio (LB-SPR)”. Between August 2022 and March 2023, a total of 1219 individuals were collected through fortnightly sampling from two fish landing centers (BFDC fishery ghat and Kaptai boat ghat) and two local markets (Banarupa Bazar and Reserve Bazar). The maximum length of *Ailia coila* was recorded at 15.7 cm in total length, with an average length of 11.1 (± 4) cm. The length-weight relationship was estimated as $W = 0.01084L^{2.599}$ ($R^2 = 0.75$). The value of shape parameter ‘ b ’ is smaller than three which indicate a negative allometric growth pattern. The estimated asymptotic length (L_∞) and length at first maturity ($L_{50\%}$) were found to be 16.84 and 10.54 cm, respectively. The LB-SPR revealed that the fishery is currently experiencing overfishing, with the spawning biomass (SPR = 32%) below the target reference point. The mean estimate of fishing mortality relative to natural mortality (F/M) was 3.07, more than three times the threshold $F/M = 1$, indicating extreme fishing mortality in this fishery. The estimated length at first capture ($SL_{50\%}$) was significantly higher than the length at first maturity ($L_{50\%}$), suggesting that the fishing is predominantly targeting mature fishes. To enhance the spawning biomass and ensure the sustainability of the fishery, it is imperative to reduce the current fishing pressure and implement appropriate management measures.

1 | Introduction

The Gangetic ailia (*Ailia coila*), locally known as ‘Kajuli’, is a freshwater small indigenous species (SIS) in Bangladesh, belonging to the family Ailiidae. It also exhibits a wide distribution across India, Pakistan, and Nepal (Wang et al. 2016; Froese and Pauly 2000). As a carnivorous species, it primarily feeds on both plants and animals and typically reaches a total length of 18 cm, although specimens up to 30 cm in

length have been recorded (Talwar and Jhingran 1991; Deori et al. 2017). Presently, this species is categorized as vulnerable in India and Bangladesh and is considered near threatened according to the IUCN Red List (Khan 2000; Mijkherjee, Praharaj, and Das 2002; Hanif et al. 2015; IUCN 2020). The man-made Kaptai Lake, known for its significant contribution to freshwater fish production in Bangladesh, also serves as a crucial habitat for the Gangetic ailia. In 2019, fish production in Kaptai Lake made up 0.24% of Bangladesh’s total

fish production. From 1983–84 to 2021–22, the fish production in the lake increased from 4057 MT to 17,937 metric ton (MT), marking a fourfold increase. However, in 2020–21, *Ailia coila* contributed 0.54% of the total fish production in the lake (BFDC, 2021). Despite its significance in the market, this indigenous fish species is facing a concerning decline in production within the lake. To prevent further decline, it is imperative to implement an effective management strategy (Sarker, Chan, and Alam 2024; Assefa, Wondie, and Enyew 2019; Raghavan et al. 2018). Developing such strategy requires essential information about the stock spawning biomass, a crucial aspect that has been lacking in previous studies conducted in Bangladesh. The absence of data on *Ailia coila*'s stock spawning biomass is a critical gap in our understanding of the stock's status. Without this knowledge, there is a risk of excessive and indiscriminate exploitation, potentially leading to short-term production gains followed by long-term decline and even the collapse of the stock (Zhai and Pauly 2019). Several length-based methods have been used for the data-poor fisheries if length data are available. These include length-based indicators (LBI; Froese 2004), length-based spawning potential ratio (LBSPR; Hordyk, Loneragan, and Prince 2015), and length-based Bayesian biomass approach (LBB; Froese et al. 2018). Among these methods, the present study applied LB-SPR method to find out the values of relative fishing mortality (F/M), selectivity, and maturity at length and estimates the spawning potential ratio (SPR) (Hordyk et al. 2015, 2016). This method assesses the impact of current fishing activities on the reproductive biomass of stocks. Length composition data and accurate biological information are required for the LB-SPR method. This method does not require knowledge of the natural mortality rate (M), but instead, the ratio of natural mortality (M) and Von Bertalanffy growth coefficient (K), (M/K) is needed (Prince, Hordyk, et al. 2015; Prince, Victor, et al. 2015). Recognizing the importance of assessing the stock in Kaptai Lake, this study aims to estimate population parameters and establish length-based reference points for *Ailia coila*. The goal is to provide valuable insights and recommendations that can guide the formulation of effective management strategies, ensuring the sustainability of this fishery.

2 | Materials and Methods

2.1 | Study Area

This study was conducted in Kaptai Lake, a man-made lake located in the Rangamati district of Bangladesh (Latitude 22°20'–23°18' N; Longitude 92°00'–92°26') with an area of 68,800 ha and the total water surface area is 58,300 ha (Galib 2009). The average water depth is about nine meters with maximum depth of 32 m where seasonal depth variation is 8.14 m (Galib 2009). Different types of gears are used to exploit the fisheries resources of this lake, that is, thela jal, bhasha jal, moshari jal, kechki jal, dharma jal, current jal, borshi etc. (Galib 2009). There are four fish landing centers in the area at Rangamati Sadar, Kaptai, Mahalchhari, and Marissha for landing and marketing of fish from the lake. However, majority of the fishes are landed in the Rangamati Sadar and Kaptai landing centers.

For the stock assessment of *Ailia coila*, length and weight data were collected from BFDC Fishery Ghat Rangamati and Kaptai (22°38' 53.0592" N; 92°11' 6.7344" E and 22°29' 51.666" N; 92°13' 12.6948" E), Banarupa Bazar (22°39' 19.4" N; 92°10' 44.2" E) and Reserve Bazar (22°39' 18.21" N; 92°11' 48.4" E) from August 2022 to March 2023 fortnightly (Figure 1).

2.2 | Data Collection

During the sampling carried out in two fish landing centers and two local markets, a total of 1219 *Ailia coila* samples were collected. Both length and weight data were collected from these samples. At the fish landing centers and the local markets, these samples were collected randomly. The total length of each sample was measured using a scientific scale with an accuracy of 0.1 cm, and the weight of each sample was determined using a digital weighing device with a precision of 0.1 g. After taking the length and weight data, these data were arranged in MS Excel spreadsheet carefully. Yearly landing data were collected from the BFDC office, Rangamati.

2.3 | Statistical Analysis

All the statistical analyses; that is, length-weight relationship, relative fishing mortality, selectivity, and spawning potential ratio (SPR) were estimated using the statistical software R (version: 4.3.3).

2.3.1 | Length-Weight Relationship

The following formula, developed by Le Cren (1951) was used to determine the length-weight relationship's where intercept ' a ' is the scaling coefficient and slope ' b ' is the shape parameter.

$$W = aL^b \quad (1)$$

where, W is the fish body's weight (g), L is the total length of fish (cm).

The value of b provides information on fish growth patterns that is, $b=3$ indicates an isometric growth, $b>3$ indicates a positive allometric and $b<3$ indicates negative allometric growth patterns (Le Cren 1951).

2.3.2 | Stock Assessment Indicators

This study used the R package LB-SPR (Length-Based Spawning Potential Ratio) proposed by Hordyk et al. (2015) to assess stock status of the *Ailia coila*. This package estimates the spawning potential ratio (SPR) as a function of the relative fishing mortality (F/M). The SPR is the natural or unfished reproductive proportion that exists in the stock under current exploitation. To know the impact of current fishing pressure on the stock's reproductive capacity, this well-established biological reference point was used (Hordyk et al. 2015; Goodyear 1993). For the fishery's sustainability, 40% SPR is recognized as the target reference point and 20% as the limit reference point (Hordyk et al. 2015).

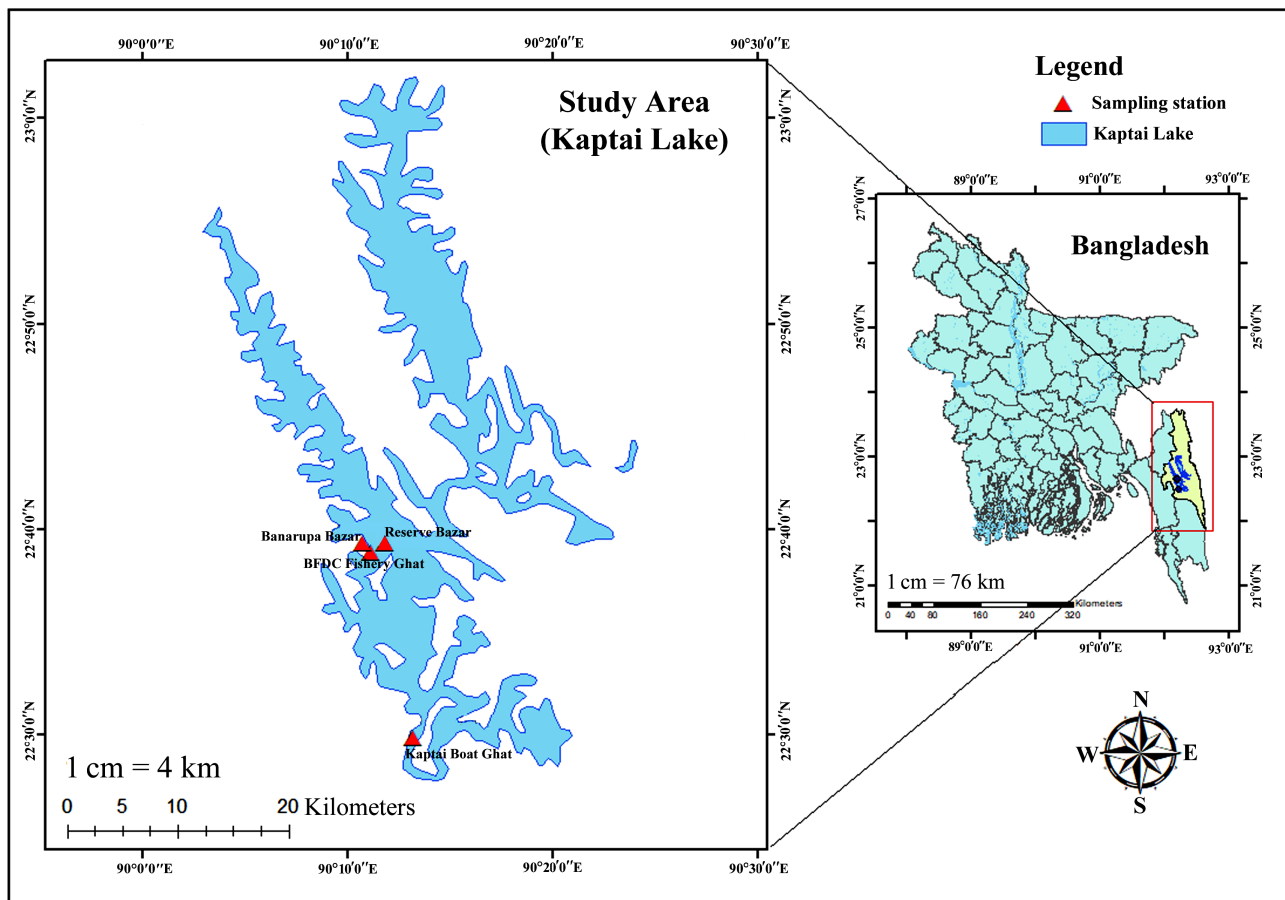


FIGURE 1 | Study area showing the locations of the sampling sites.

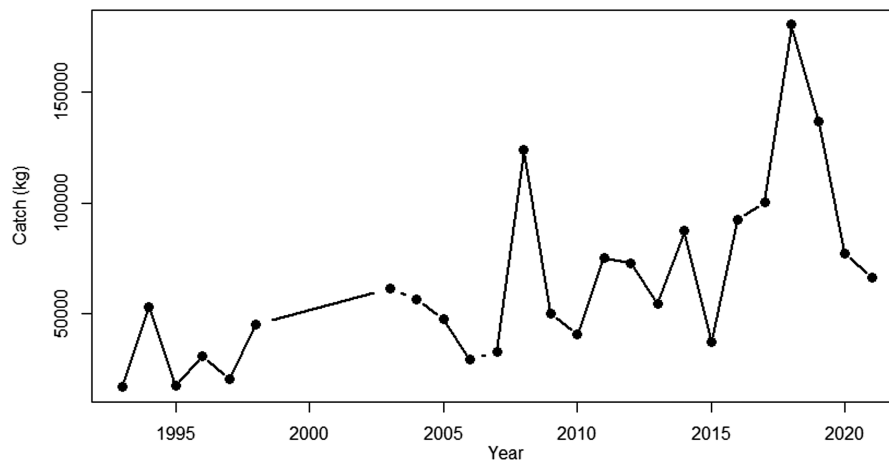


FIGURE 2 | Trends of the annual landing of *Ailia coila* in Kaptai Lake, Rangamati, Bangladesh from 1993–2021.

2.3.3 | Length-Based Spawning Potential Ratio (LB-SPR)

The LB-SPR method estimates the spawning potential ratio (SPR) that has a solid theoretical foundation and has been tested widely and it is benchmarked thoroughly in the literature (Dowling et al. 2019; Pons et al. 2019; Hordyk et al. 2015; Prince, Hordyk, et al. 2015; Prince, Victor, et al. 2015; Beverton and Holt 1959; Rudd and Thorson 2017; Babcock, Tewfik, and Burns-Perez 2018; Chong et al. 2020). The model assumes that the length structure and the SPR of an exploited stock are a

function of relative fishing mortality (F/M) and life history ratios, M/K (Prince, Hordyk, et al. 2015; Prince, Victor, et al. 2015). Therefore, the LB-SPR model also needs the following input parameters in addition to the length distribution data: (i) the M/K ratio; (ii) the asymptotic length (L_{∞}); (iii) the lengths at which 50% and 95% of a fish population are mature (Prince, Hordyk, et al. 2015; Prince, Victor, et al. 2015).

With the calculation of the SPR, this model also estimates the relative fishing mortality (F/M) and the length of the first capture of 50% ($SL_{50\%}$) and 95% ($SL_{95\%}$) of the population using

the maximum likelihood methods (Hordyk et al. 2016; Prince, Hordyk, et al. 2015; Prince, Victor, et al. 2015). Unlike other length-based stock assessment methods, this model works under several assumptions, including asymptotic selectivity, growth is accurately described by the Von Bertalanffy equation, length-at-age is normally distributed, constant natural mortality rates across adult age groups, and consistent growth rates across cohorts of the stock (Prince, Hordyk, et al. 2015; Prince, Victor, et al. 2015). For the estimation of SPR, the value of L_{∞} (asymptotic length) was calculated from the following formula proposed by Pauly and Munro (1984).

$$L_{\infty} = L_{\max} / 0.95 \tag{2}$$

where, $L_{\max}$ is the observed maximum length of the fish.

The following equation, developed by Froese and Binohlan (2000) was used to estimate the length at 50% sexual maturity ($L_{50\%}$) from estimated L_{∞} .

$$\log L_{50\%} = 0.8979 \log L_{\infty} - 0.0782 \tag{3}$$

Another equation, proposed by Zhang et al. (2023), was used to estimate the length at 95% sexual maturity ($L_{95\%}$).

$$L_{95\%} = 1.1 \times L_{50\%} \tag{4}$$

TABLE 1 | The number of samples collected monthly from August 2022 to March 2023.

Month	August	September	October	November	December	January	February	March
Number of samples	153	152	153	152	151	153	151	154

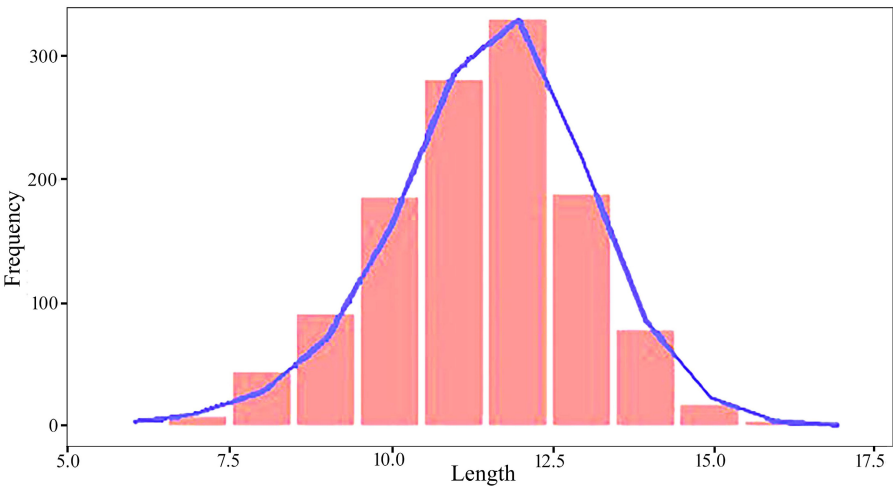


FIGURE 3 | The bar chart shows the length–frequency, and the curve shows the unimodal distribution of the length data of the *Ailia coila* during the study period.

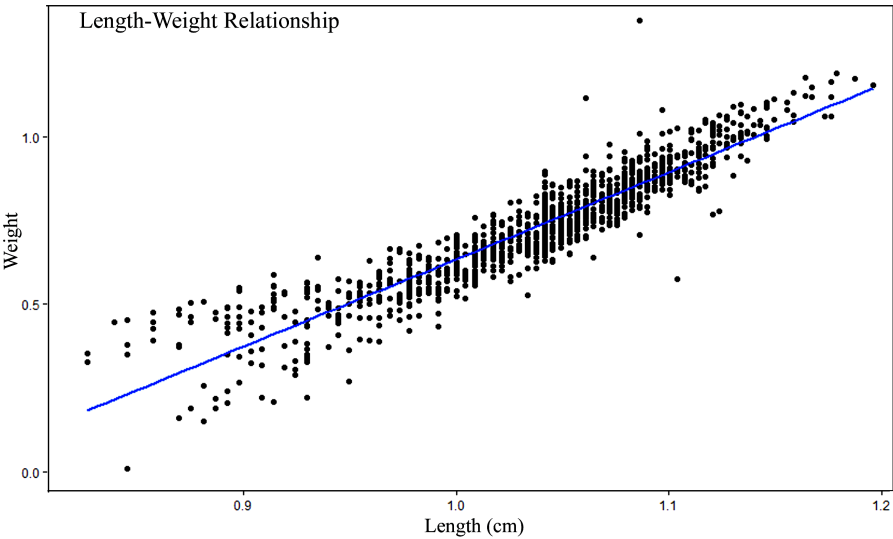


FIGURE 4 | Length-weight relationship of the *Ailia coila* from the kaptai lake, Rangamati, Bangladesh.

3 | Results

3.1 | Trends in Landing

After evaluating the yearly landing catch data collected from the Bangladesh Fisheries Development Corporation (BFDC), the study observed a declining trend for *Ailia coila* at present (Figure 2). The landing gradually shows ups and downs during the years 1993–2007, then suddenly increased in 2008. Though it was again showed fluctuations in the years between 2009 and 2015, the catches had increased gradually from the years 2016 to 2018. Nonetheless, catches once again showed a decreasing trend between 2019 and 2021. The highest catch was recorded in 2017–18, reaching 180.56 MT, while the average catch for the last 5 years was 112.27 MT. However, there was a notable decrease in catch in 2021, with only 66.22 MT recorded.

3.2 | Length Frequency Distributions

A total of 1219 individuals of the *Ailia coila* were measured for total length (TL) during study period (Table 1). Due to 4 months fishing ban, collecting data was not possible from the month of April to July. It was observed that the length-frequency distribution was unimodal, with one distinct peak, and the modal size ranged from 9.5 to 13.5 cm (Figure 3). During the sampling period, no difference in size either temporally or spatially was observed.

3.3 | Length-Weight Relationship

The estimated value of a (Scaling coefficient) and b (Shape parameter) were obtained from total length (TL) and total weight

(W) data using the length-weight relationship equation where $a=0.01084$ and $b=2.599$ and thus the length-weight relationship was $W=0.01084L^{2.599}$ ($R^2=0.75$), and the value of b is smaller than 3 ($b<3$); indicates the negative allometric growth pattern of the species (Figure 4).

3.4 | Spawning Potential Ratio (SPR)

3.4.1 | Estimation of Life History Ratio (LHR) and Size of Maturity

Estimation of the two LHR (L_{∞} and M/K) is needed for the LB-SPR assessment. L_{∞} was estimated using Equation (2). The M/K ratio was set to 1.5 (Froese et al. 2018). We calculated the size probability at which 50% ($L_{50\%}$) and 95% ($L_{95\%}$) of maturity occurred using Equations (3) and (4) (Table 2).

3.4.2 | Model Fitting to the Length Distribution Data

The calculated mean length was 11.1 (± 4) cm. The estimated growth curve by LB-SPR fits well with the distribution of our data and a normal distribution curve was produced (Figure 5).

3.4.3 | Length Selectivity and Maturity

From the automated LB-SPR process in the R studio (R core team 2018), the mean estimates of 50% ($SL_{50\%}$) and 95% ($SL_{95\%}$) selectivity are 11.4 cm and 14.17 cm respectively (Table 3). The mean estimate of F/M was 3.07 which is more than thrice the threshold $F/M=1$. The selectivity and maturity ogive curve demonstrated that the length at maturity is higher than the length at first capture (Figure 6).

3.4.4 | Spawning Potential Ratio (SPR)

The estimated value of SPR for the year 2022 was 32% that is higher than the limit reference point (LRP) of 20% (Figure 7 and Table 3). However, this value is lower than the target reference point (40%), indicating that the stock is overfished.

TABLE 2 | Input parameters for LB-SPR analysis.

Parameters	Value
L_{∞}	16.84 cm
M/K	1.5
Length at 50% maturity ($L_{50\%}$)	10.54 cm
Length at 95% maturity ($L_{95\%}$)	11.6 cm

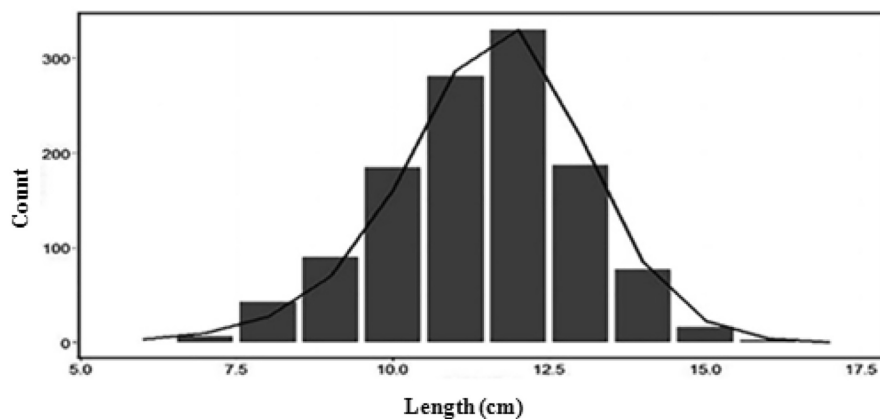


FIGURE 5 | Length-frequency distribution of the *Ailia coila* stock, and the solid black line shows the predicted fished size composition from the fitted LB-SPR model.

3.4.5 | Existing vs. Expected Size Composition at the Target SPR (0.40)

The length classes that should be targeted to maintain the threshold level (SPR=40%) are highlighted by red colour in Figure 8. Though, the current selectivity of the fishery mainly targeted the matured and larger fishes, a significant amount of juvenile fishes still be targeted.

TABLE 3 | Outputs form LB-SPR analysis.

Parameters	Values
SL _{50%}	11.4 cm
SL _{95%}	14.17 cm
F/M	3.07
SPR	0.32

4 | Discussion

Despite the significant importance of Kaptai Lake fisheries, they have not received adequate attention from relevant regulatory authorities or research endeavours. Effective management policies for sustainable fisheries depend on addressing all biological reference points associated with the fishery (Alam et al. 2021). However, in Bangladesh, data availability is minimal, and the available data are often incomplete and misreported. Consequently, policymakers lack the necessary high-quality data for proper stock assessment, hindering the development of long-term management plans for this fishery (Alam, Liu, Monwar, et al. 2022; Alam, Liu, Schneider, et al. 2022). Moreover, from the inception of the time series, the yearly catch data of Gangetic ailia shows a dynamic trend until it peaked in 2018. However, the trend sharply declines in the subsequent years, reaching its lowest in 2021. In fact, the catch in 2021 decreased by more than 63% compared to the peak catch of 2018, with over half of the catch diminishing within just three years. This rapid and

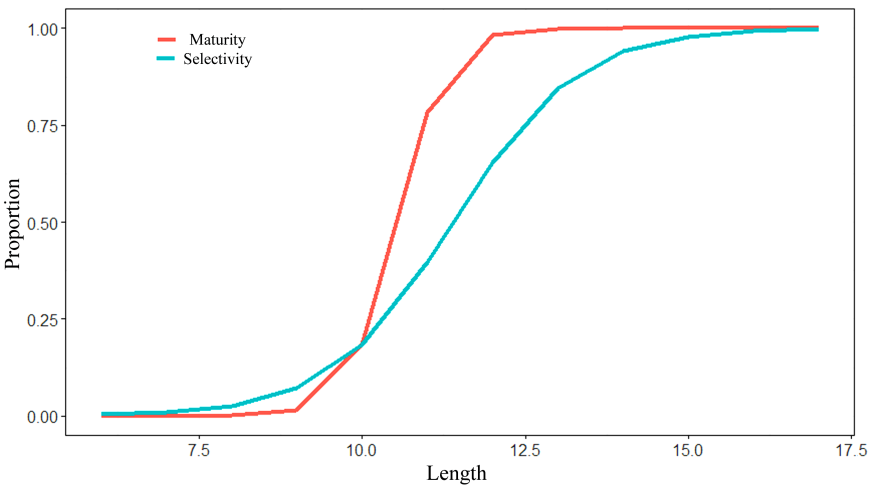


FIGURE 6 | Maturity and selectivity curve from the fitted LB-SPR model when $L_{50\%} = 10.54$ cm and $L_{95\%} = 11.60$ cm.

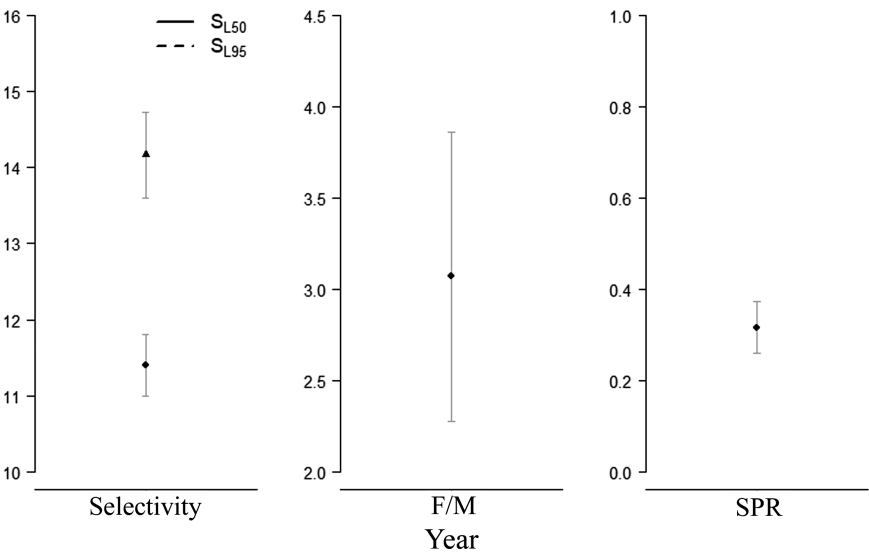


FIGURE 7 | The distribution of mean selectivity parameters ($SL_{50\%}$ and $SL_{95\%}$), fishing mortality to natural mortality (F/M) and spawning potential ratio (40%).

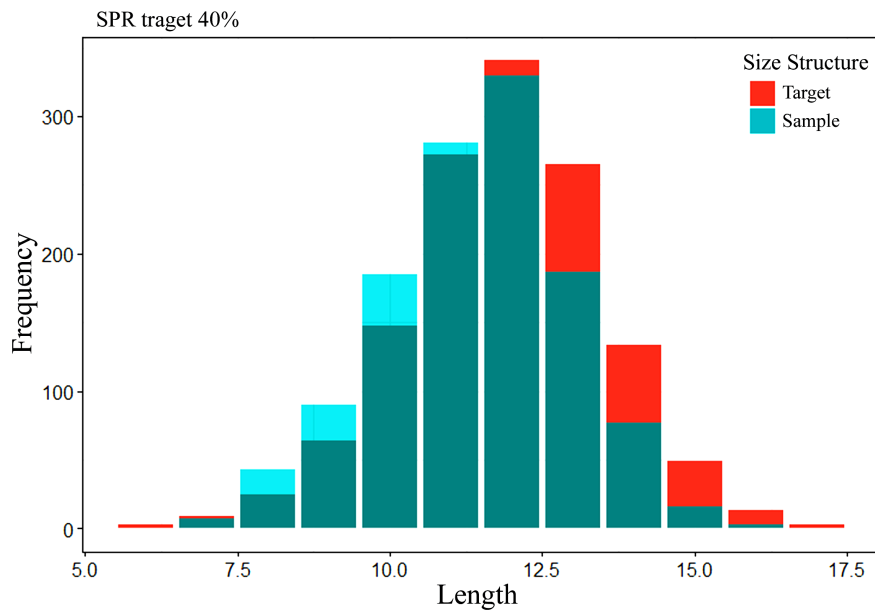


FIGURE 8 | Observed size data against an expected size composition at a target SPR (40%).

TABLE 4 | Estimates of LWR coefficient (b) from different studies.

Location	LWR coefficient b	Growth pattern	Reference
Kaptai Lake	2.599	Negative allometric	This study
Padma River	3	Isometric	Hossain (2010)
Ganga River, India	3.13	Positive allometric	Ray et al. (2022)

significant drop has raised serious concerns among stakeholders regarding the current stock status of this species in the lake. To address this challenge, this study employed the LBSPR method, which requires minimal data (length data only) to estimate the fishery's reference points. These reference points will be crucial in formulating effective management policies for the sustainable management of this fishery in Kaptai Lake.

Length-weight relationships (LWRs) play a crucial role in stock assessment and are important in converting growth-in-length equations to growth-in-weight (Mondal et al. 2024). This aids in estimating the stock status and understanding the life histories of various fish species (Binohlan and Pauly 1998; Ahmed and Saha 1996; Moutopoulos and Stergiou 2002; Nile et al. 2013; Mortuza and Almisned 2013). Particularly for threatened fish species like *Ailia coila*, LWRs provide essential growth and condition information crucial for management and conservation efforts (Sarkar, Deepak, and Negi 2009; Muchlisin, Musman, and Azizah 2010). Research on the length-weight relationship of *Ailia coila* in Kaptai Lake is lacking, but studies in other water bodies in Bangladesh and India show conflicting results (Table 4). Hossain (2010) found an isometric growth pattern in the Padma River, Bangladesh, while Ray et al. (2022) reported positive allometric growth in the Ganga River, India. These variations in growth patterns across different locations highlight potential differences in environmental conditions and ecological factors influencing the growth of *Ailia coila* (Alam et al. 2021).

Spawning potential ratio (SPR) provides valuable insights into the reproductive potential of the fish stock. Even with limited data, understanding the proportion of mature fish capable of spawning is crucial for estimating the stock's ability to replenish itself (Hordyk et al. 2015). Determining the value of SPR is essential for devising effective fish stock management strategies aimed at ensuring sustainable fisheries. In this study, we used the LB-SPR model developed by Hordyk et al. (2015) to evaluate the spawning biomass of *Ailia coila* in the water of Kaptai Lake. The goal of fisheries management is to maintain fish stocks at sustainable levels to secure future fishery resources. By comparing the current SPR to reference points, fisheries managers can gauge whether the stock is being harvested sustainably or if it is being overexploited. In the absence of comprehensive data, SPR can guide management decisions by helping set appropriate catch limits and fishing regulations. It provides a basis for determining if fishing pressure needs to be reduced to ensure the long-term sustainability of the fishery (Hordyk et al. 2015; Alam, Liu, Monwar, et al. 2022; Alam, Liu, Schneider, et al. 2022). Our analysis revealed that the fishery is experiencing extensive fishing mortality ($F/M = 3.07$), with the present SPR (0.32) falling below the target reference point. Despite the fact that the length at first capture is higher than the length at first maturity, the findings of this study indicate overexploitation, likely attributed to indiscriminate catches of various life stages, resulting in a scarcity of targeted length classes required to maintain the threshold SPR level (0.40).

The asymptotic length (L_{∞}) is a key parameter in fisheries biology and stock assessment, aiding in understanding fish population dynamics. Our study in Kaptai Lake, Bangladesh, found a significantly lower L_{∞} for *Ailia coila* compared to studies in the Brahmaputra (26.80 cm) and Ganga Rivers (18.9 cm) by Gogoi et al. (2021) and Ray et al. (2022) respectively. While variations in L_{∞} can arise from factors like habitat and environmental conditions, fishing pressure remains a primary factor in reducing L_{∞} over time. Prolonged overfishing selectively removes larger, older individuals, leading to a shift in size structure towards smaller, younger individuals, consequently yielding lower values for L_{∞} (Alam et al. 2021). Overfishing may be attributed to poor fisheries management, unsustainable fishing practices, IUU (Illegal, Unreported, and Unregulated) fishing activities, bycatch, and the use of harmful fishing gear with illegal mesh size. These findings underscore the urgent need for reducing current fishing mortality and implementing proper management measures to ensure the sustainability of the fishery.

4.1 | Management Recommendation

The research findings highlight the urgency of implementing management measures to address growth overfishing of *Ailia coila* in Kaptai Lake. Regulatory actions should focus on protecting juveniles by mandating the use of appropriate mesh sizes in fishing nets (Vahabnezhad et al. 2021). Enforcing size-based regulations, with the length at first capture exceeding the measured length at first sexual maturity ($L_{50\%}$) of 10.54 cm, is crucial (Alam, Liu, Monwar, et al. 2022; Alam, Liu, Schneider, et al. 2022). Strict prohibitions on catching fish below this length are essential to allow immature individuals to spawn and attain optimal size. Despite existing bans on fishing from April to mid-August, still illegal fishing persists due to inadequate monitoring and limited alternative livelihood opportunities. To combat this, comprehensive monitoring mechanisms should be established, coupled with government assistance programs to support fishermen and other lake-dependent livelihoods during the closed season (Rayhan et al. 2021; Shalehin, Naher, and Galib 2022). Additionally, ongoing evaluation and refinement of management strategies through formal monitoring processes are essential to ensure their effectiveness and the long-term sustainability of *Ailia coila* populations in Kaptai Lake.

5 | Conclusions

The fisheries resources of Kaptai Lake has been experiencing excessive fishing pressure which made the majority of fish stocks of this lake vulnerable which is evident from an continuous reduction of catches. There, necessary steps need to adopt to save those fisheries. Given the unavailability of necessary data for conventional stock assessment, this study employed a data-poor length-based stock assessment using LB SPR to reveal the current stock status of an indigenous fish species, *Ailia coila*. The findings reveal that the species is overfished with spawning biomass below target levels and fishing mortality rates exceeding sustainable thresholds. The estimated length at first capture and maturity suggests a selective pressure on

mature individuals, exacerbating the depletion of spawning potential. To safeguard the future sustainability of the fishery and restore spawning biomass, urgent interventions are required, including reducing fishing pressure and implementing effective management strategies. Addressing these challenges is crucial to conserve *Ailia coila* populations in Kaptai Lake and ensuring continued socio-economic benefits from this valuable freshwater resource.

Author Contributions

A.R.M.: data collection, methodology, data analysis, and writing, M.R.S.: conceptualization, methodology, data analysis, visualization, writing and editing, M.S.A.: conceptualization, methodology, data analysis, visualization, reviewing, editing, and supervision. All authors have read and agreed to the published version of the manuscript.

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Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be available on request to corresponding author.

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